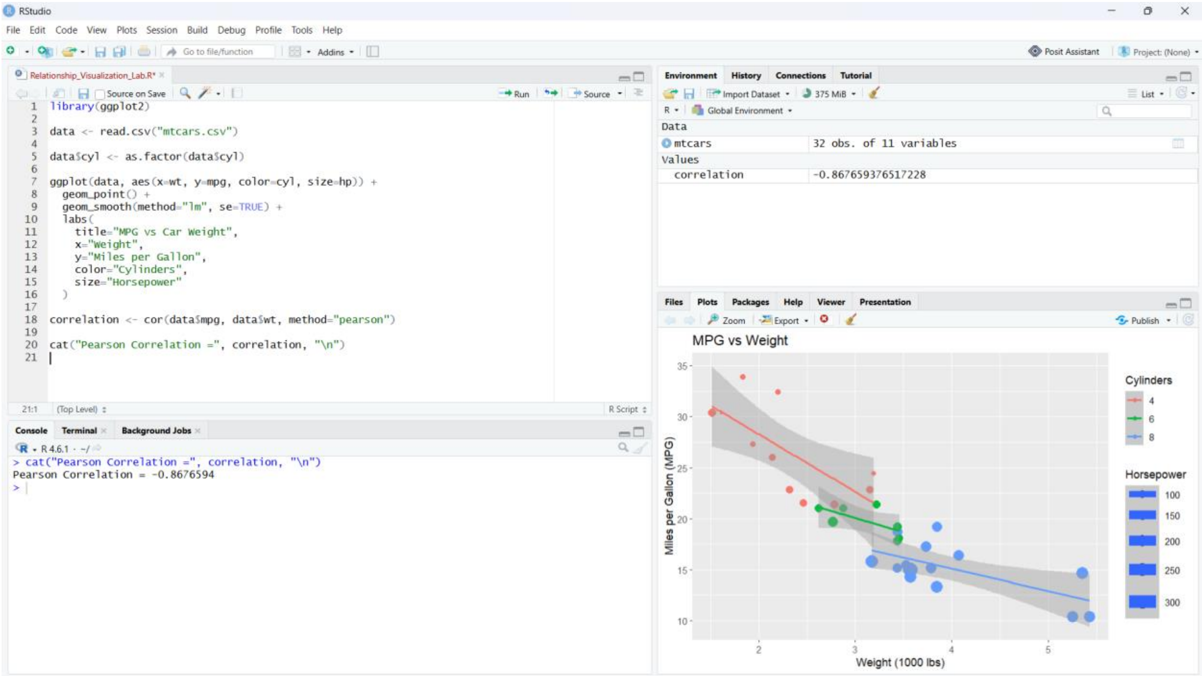
QUESTION 1

Code:

```
library(ggplot2)
data <- read.csv("mtcars.csv")
data$cyl <- as.factor(data$cyl)
ggplot(data, aes(x=wt, y=mpg, color=cyl, size=hp)) +
  geom_point() +
  geom_smooth(method="lm", se=TRUE) +
  labs(
    title="MPG vs Car Weight",
    x="Weight",
    y="Miles per Gallon",
    color="Cylinders",
    size="Horsepower"
  )
correlation <- cor(data$mpg, data$wt, method="pearson")
cat("Pearson Correlation =", correlation, "\n")
```

Output:



Interpretation

The relationship between **mpg and wt is negative**, meaning heavier cars generally have lower fuel efficiency. The correlation is strong and negative, showing that weight is an important factor associated with MPG.

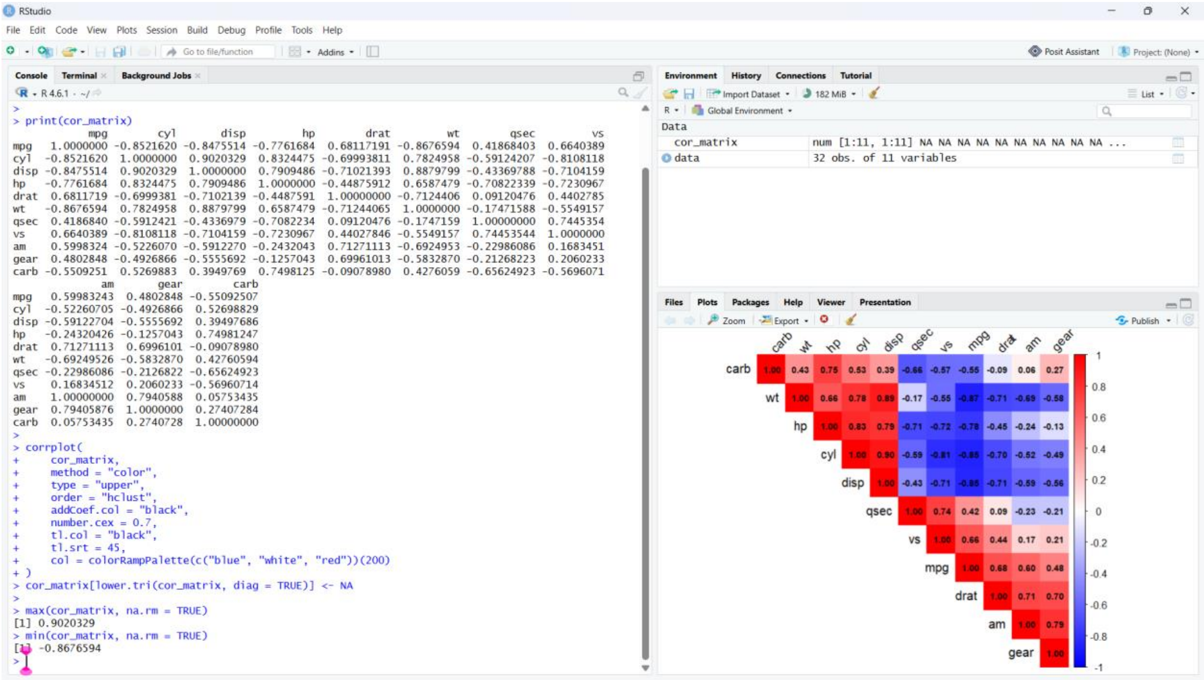
QUESTION 2

Code:

```
library(corrplot)

data <- mtcars

cor_matrix <- cor(data)

print(cor_matrix)

corrplot(

  cor_matrix,

  method = "color",

  type = "upper",

  order = "hclust",

  addCoef.col = "black",

  number.cex = 0.7,

  tl.col = "black",
```

```
  tl.srt = 45,
  col = colorRampPalette(c("blue", "white", "red"))(200)
)
```

Output:



Interpretation

- The correlogram shows the correlation between all numeric variables in mtcars.
- **Strongest positive:** cyl and disp ≈ **0.90**.
- **Strongest negative:** mpg and wt ≈ **-0.87**.
- **Red** indicates positive correlation, while **blue** indicates negative correlation.
- Hierarchical clustering groups variables with similar correlation patterns.

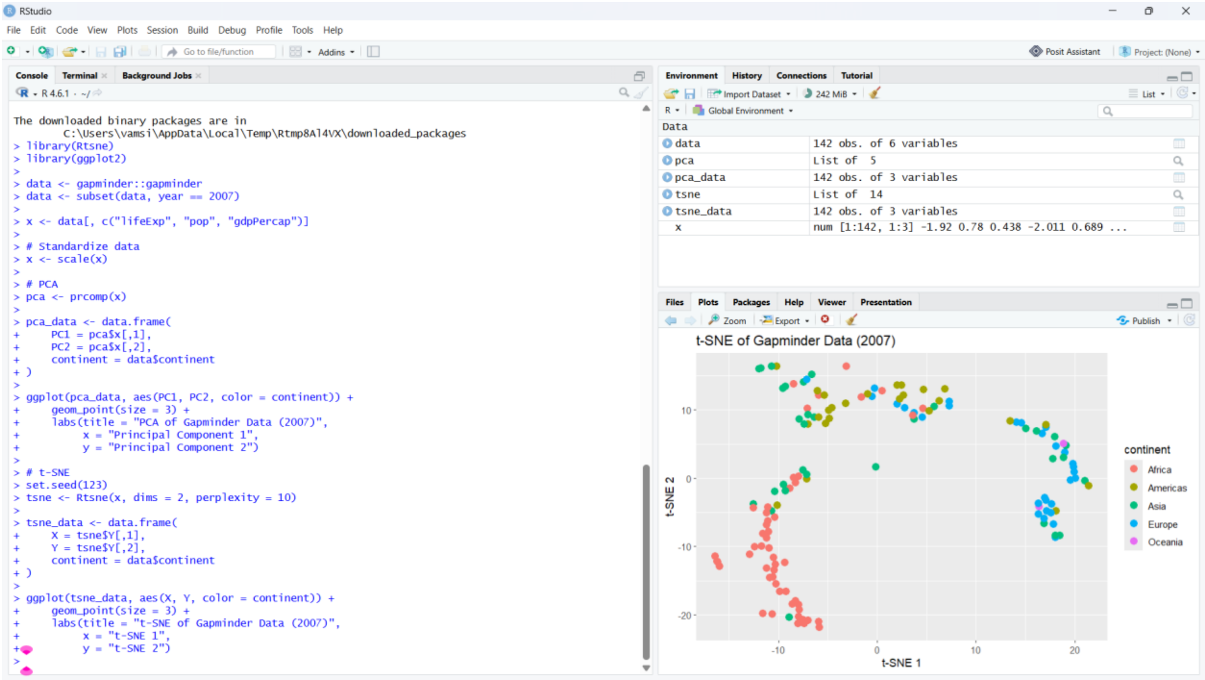
QUESTION 3

Code:

```
install.packages("Rtsne")
library(Rtsne)
library(ggplot2)
data <- gapminder::gapminder
data <- subset(data, year == 2007)
```

```r
x <- data[, c("lifeExp", "pop", "gdpPercap")]
# Standardize data
x <- scale(x)
# PCA
pca <- prcomp(x)
pca_data <- data.frame(
  PC1 = pca$x[,1],
  PC2 = pca$x[,2],
  continent = data$continent
)
ggplot(pca_data, aes(PC1, PC2, color = continent)) +
  geom_point(size = 3) +
  labs(title = "PCA of Gapminder Data (2007)",
       x = "Principal Component 1",
       y = "Principal Component 2")
# t-SNE
set.seed(123)
tsne <- Rtsne(x, dims = 2, perplexity = 10)
tsne_data <- data.frame(
  X = tsne$Y[,1],
  Y = tsne$Y[,2],
  continent = data$continent
)
ggplot(tsne_data, aes(X, Y, color = continent)) +
  geom_point(size = 3) +
  labs(title = "t-SNE of Gapminder Data (2007)",
       x = "t-SNE 1",
       y = "t-SNE 2")
```

Output:



Interpretation

- **PCA** reduces the data while preserving maximum overall variance.
- **t-SNE** is better at showing local clusters and similarities.
- PCA gives a more interpretable global structure, while t-SNE may show clearer clusters.
- Both methods reduce the original variables to **2 dimensions**.

QUESTION 4

Code:

library(ggplot2)

library(gapminder)

data <- subset(gapminder, year == 2007)

ggplot(data, aes(x = gdpPercap, y = lifeExp,

size = pop, color = continent)) +

geom_point(alpha = 0.7) +

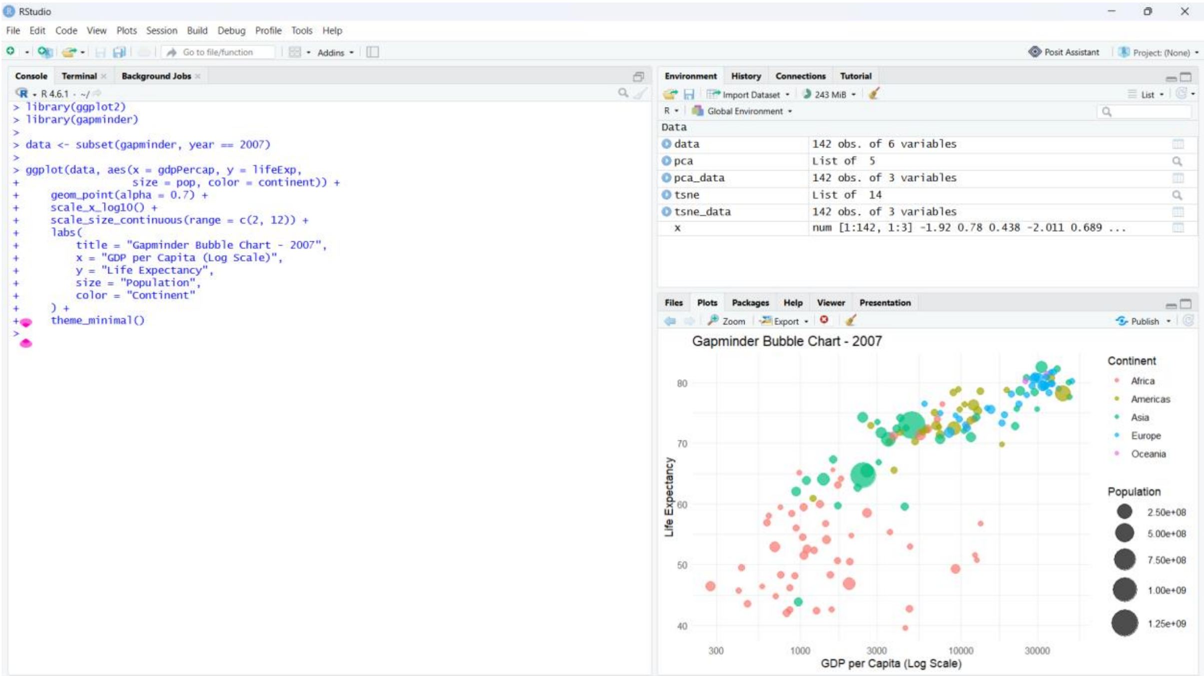
scale_x_log10() +

scale_size_continuous(range = c(2, 12)) +

labs(

```
  title = "Gapminder Bubble Chart - 2007",
  x = "GDP per Capita (Log Scale)",
  y = "Life Expectancy",
  size = "Population",
  color = "Continent"
) +
theme_minimal()
```

Output:



Interpretation

- The chart shows **GDP per capita vs. life expectancy** in 2007.
- Bubble size represents **population**, while color represents **continent**.
- Countries with higher GDP generally have **higher life expectancy**.
- **Asia** contains some very large bubbles due to highly populated countries such as China and India.
- **Africa** forms a visible lower-life-expectancy cluster.

QUESTION 5

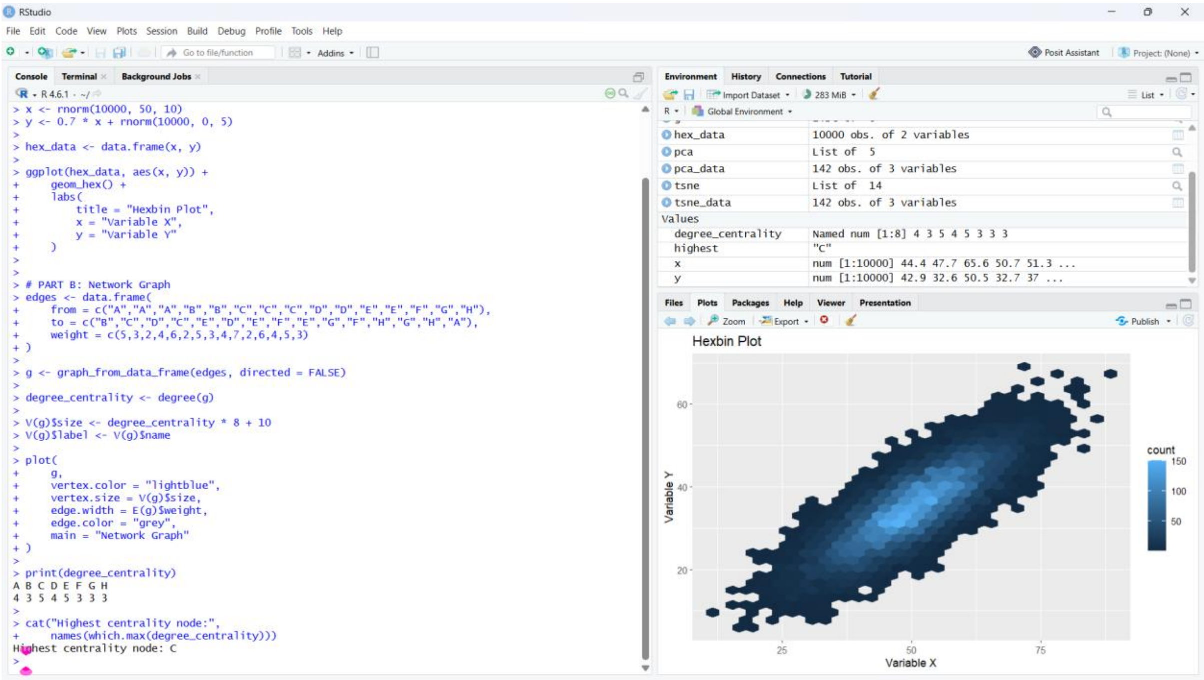
Code:

```
library(ggplot2)
library(igraph)
# PART A: Hexbin Plot
set.seed(123)
x <- rnorm(10000, 50, 10)
y <- 0.7 * x + rnorm(10000, 0, 5)
hex_data <- data.frame(x, y)
ggplot(hex_data, aes(x, y)) +
  geom_hex() +
  labs(
    title = "Hexbin Plot",
    x = "Variable X",
    y = "Variable Y"
  )
# PART B: Network Graph
edges <- data.frame(
  from = c("A","A","A","B","B","C","C","C","D","D","E","E","F","G","H"),
  to = c("B","C","D","C","E","D","E","F","E","G","F","H","G","H","A"),
  weight = c(5,3,2,4,6,2,5,3,4,7,2,6,4,5,3)
)
g <- graph_from_data_frame(edges, directed = FALSE)
degree_centrality <- degree(g)
V(g)$size <- degree_centrality * 8 + 10
V(g)$label <- V(g)$name
plot(
  g,
  vertex.color = "lightblue",
  vertex.size = V(g)$size,
  edge.width = E(g)$weight,
  edge.color = "grey",
```

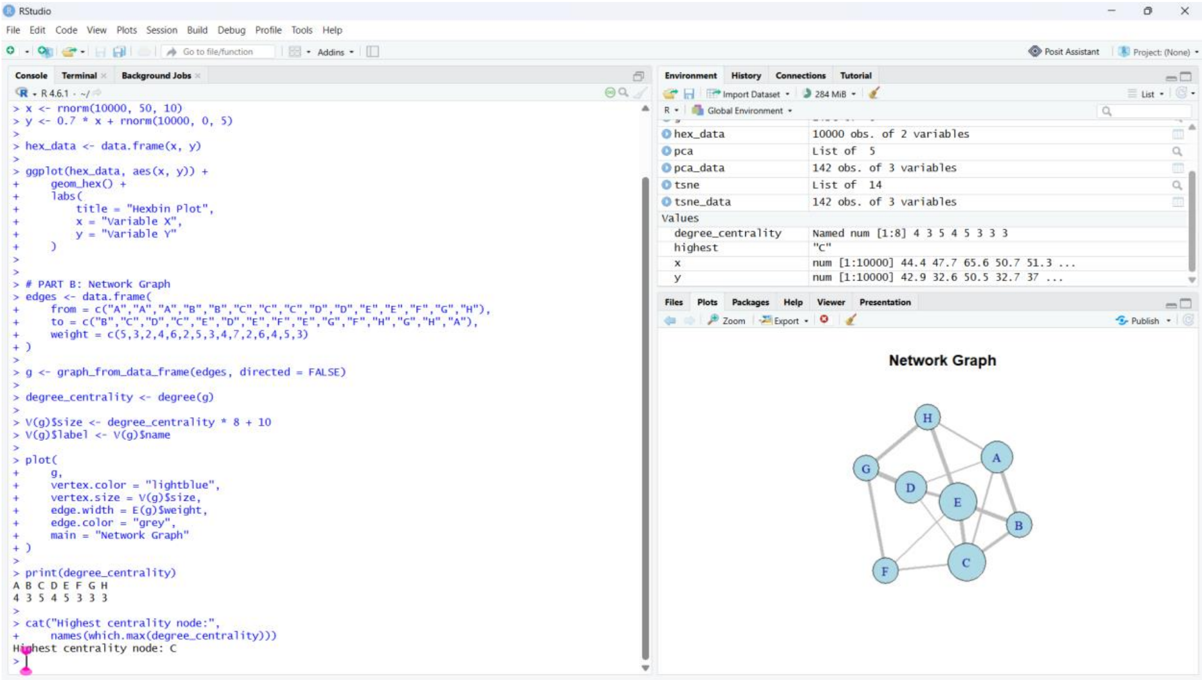
```
  main = "Network Graph"
)
print(degree_centrality)
cat("Highest centrality node:",
    names(which.max(degree_centrality)))
```

Output:

a) Hexbin Plot



b) Network Graph



Interpretation

Hexbin: Shows the density of 10,000 observations and reduces overplotting. It clearly shows a positive relationship between X and Y.

Network Graph: Larger nodes have higher degree centrality, while thicker edges represent greater weights. The most connected node acts as a **hub** in the network.